

Framework Assessment

Four Benchmarking Frameworks for
Deep Tech Commercialization at Early TRL

Milestone 1, Companion to the Literature Map

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Executive Summary

This document presents a detailed assessment of four deep tech commercialization frameworks selected as initial benchmarking candidates for the NCCR Muoniverse thesis project. Each framework is the primary reference point for a different stakeholder in the Swiss deep tech ecosystem: the RCA/Imperial College module is what NCCR Muoniverse currently references; the Commercial Readiness Level (CRL) framework is widely cited in the deep tech discourse as the leading assessment instrument; the Innosuisse coaching methodology is the native Swiss support framework; and the European Innovation Council (EIC) represents the EU-level funding architecture that Swiss deep tech ventures can now access.

Each framework is assessed across three layers: primary documentation (what the framework is), academic evidence (what supports it), and fit for NCCR Muoniverse (how relevant it is for TRL 1–3 physics ventures in Switzerland). Throughout this document, bracketed numbers [1], [2], etc. refer to entries in the References section, where full source URLs are provided.

The headline finding across all four assessments is that every framework has a structural gap at TRL 1–3.

The RCA/Imperial modules are designed for TRL 4–6 technologies [1][2][5]. The CRL framework is conceptually applicable at all levels but operationally thin at low levels, with no published assessment methodology [11][12][13]. Innosuisse is optimized for the TRL 4–7 'valley of death' and its closest early-stage instrument (BRIDGE Proof of Concept) provides only 12 months of support [25][26]. The EIC's flagship Accelerator requires TRL 5 or above [33][34]. The space where NCCR Muoniverse operates, fundamental physics at TRL 1–3, is precisely where no existing framework was designed to work.

This gap is the central finding that motivates the thesis. If no existing framework adequately serves the earliest stages of deep tech commercialization, then this thesis contribution lies in identifying what support mechanisms are needed in this unaddressed space and how existing frameworks can be adapted or combined to reach it.

Criterion	RCA/Imperial	CRL (Abbas)	Innosuisse	EIC Horizon Europe
TRL stage designed for	4–6 [1][5]	1–9 (all) [11]	4–7 [25]	1–8 (pipeline) [33]
Evidence base	Pedagogical [8]	Proprietary consultancy [11][13]	KOF-verified causal [28][29]	Self-reported impact [36]
Swiss fit	Low (UK institution)	N/A (assessment tool)	Native (Swiss agency) [24]	High (since Jan 2025) [41]
TRL 1–3 applicability	Weak [5]	Conceptual only [16]	Moderate (BRIDGE PoC) [26]	Moderate (Pathfinder) [33]
Independently validated	No	No	Yes (KOF DiD study) [29]	No

Legend: The 'TRL stage designed for' row indicates the technology readiness range that each framework's instruments were built to serve. TRL 1–3 means the technology exists as fundamental research or proof of concept; TRL 4–6 means a prototype exists and is being tested in relevant environments; TRL 7–9 means the technology is near or at market deployment. NCCR Muoniverse operates at TRL 1–3, the least served range across all four frameworks.

Framework 1: RCA / Imperial College Deep Tech Acceleration Module

1.1 What the Module Actually Is

The label 'RCA/Imperial Deep Tech Acceleration Module' does not refer to a single standalone program. It is a cluster of interrelated educational and venture-building activities co-developed by Imperial College London, primarily through its Business School, and the Royal College of Art (RCA), one of the world's leading art and design universities [1][2]. Understanding what this cluster actually contains is essential because the name suggests a structured commercialization program, while the reality is more nuanced.

Four distinct components comprise the cluster. First, the Imperial Innovation Challenge is a one-week intensive module embedded in Imperial's Full-Time MBA curriculum. Student teams of four to six people work with real Imperial technologies, drawn from engineering, medicine, and chemistry departments, that are described as 'still under development.' The teams apply business frameworks, design speculative applications for the technologies, and pitch their proposals to venture capital and industry judges in a format modeled on the television show *Dragons' Den* [4]. This is a pedagogical exercise, not a spin-off creation mechanism.

Second, the Wicked Module is a one-month program for Imperial Business School students, co-taught with RCA faculty, that applies 'Wicked Acceleration' frameworks to complex systems-level problems [3]. Steve Blank, the creator of the Lean Startup methodology, co-taught the module in 2023 and published a detailed account describing the five-week curriculum: stakeholder mapping, assumption mapping, experimentation design, and the 'Wicked Canvas' (an extension of Osterwalder's Business Model Canvas that adds unintended consequences and stakeholder-specific sub-problems). Each week, student teams were required to speak with 10 external stakeholders [6].

Third, Wicked Acceleration Labs (WAL) is the joint Imperial-RCA research and toolkit platform that provides the intellectual framework underlying both modules. WAL operates with three pillars: Envision (collective imagination of desirable futures), Orchestrate (mobilize resources and ecosystems), and Accelerate (implementation) [2]. Its toolkit includes the Deep-Tech Acceleration Sprint, the Wicked Canvas, Stakeholder Maps, Systems Maps, and an 'Acupuncture Map' for regional system intervention [2][3].

Fourth, and quite separately from the MBA modules, DT Prime is Imperial College's cohort-based venture builder for deep tech and life science spinouts. It started in January 2022, runs 6–9 months per cohort, and provides non-dilutive funding of £50,000 to £150,000 per phase (cumulative up to £500,000) [7]. DT Prime targets Imperial College staff and researchers only. Five cohorts have been run through 2025–2026. The program focuses on four areas: therapeutics and biotech; medical technology and digital health; defence and IT hardware including photonics, quantum, and neuromorphic systems; and resilient and low-carbon technologies [7].

1.2 The Methodology: 'Flipping' Lean Startup for Deep Tech

The most distinctive aspect of the WAL methodology is what Mike Pinder (WAL co-founder and former IDEO design director) describes as 'flipping Lean Startup logic on its head' [5]. Standard Lean Startup methodology, as developed by Steve Blank and Eric Ries, follows a customer-discovery-first logic: start with a customer problem, build a minimum viable product, test it with customers, and pivot based on feedback. This works well when a market exists. But deep tech ventures typically begin with a technology capability, not a market problem, and in many cases, the market for the technology does not yet exist at all.

The WAL methodology inverts this logic. Teams start with the emerging technology capability, speculate on near-future applications using speculative design techniques borrowed from RCA's design tradition, and only then seek lead and extreme users to validate these speculative applications [5]. This technology-push approach is specifically designed for situations where no customer exists to discover because the technology is too early to demonstrate. Steve Blank himself has endorsed this as a legitimate evolution of Lean Startup for deep tech contexts [6].

1.3 What the RCA Brings

The Royal College of Art is consistently ranked as the world's number one art and design university (QS World University Rankings) [1]. Its contribution to the deep tech acceleration cluster is not decorative, it is methodological. RCA brings service design expertise, speculative design methods (imagining near-future scenarios for emerging technologies), user-centered design approaches, and visualization capabilities that help make complex scientific concepts legible to non-technical audiences including investors and potential customers. The Imperial-RCA partnership is built on a 40-year joint MA/MSc programme in Innovation Design Engineering that combines design, engineering, science, and enterprise [1][2].

1.4 Evidence Base and Academic Standing

The Institute for Deep Tech Entrepreneurship at Imperial College was founded in November 2022 with a stated ambition to 'prime 160 pre-commercial Deep Tech moonshots' over 10 years, targeting £50 million in philanthropic funding [8]. The Institute's academic lead is Professor Ramana Nanda, formerly of Harvard Business School, and its chair is Professor Erkkö Autio, one of the world's most cited innovation scholars (h-index 92, approximately 70,000 citations) [9][10].

However, no peer-reviewed outcome evaluation of the module's commercialization effectiveness exists yet [8]. The Innovation Challenge and Wicked Module are pedagogical exercises whose outputs are student presentations, not actual spin-offs. DT Prime has run five cohorts and supported Imperial researchers with non-dilutive funding, but specific spin-out outcomes are not publicly enumerated [7]. The WAL methodology has been validated through partnerships with organizations including the BBC, the UK Ministry of Defence,

the UNDP, and the World Economic Forum [2], but these are consultancy engagements, not controlled evaluations.

The broader UK policy context is relevant. The Tracey/Williamson Independent Review of University Spin-out Companies (November 2023), commissioned by the UK Chancellor of the Exchequer, made 11 recommendations including that universities should take equity stakes of 10–25% (down from as high as 50% previously) [14]. Following the review, average UK university equity stakes dropped from 22% in 2023 to 16% in 2024 , a decade low [15]. Imperial College ranked 10th in Europe for spinout value creation according to the Spotlight on Spinouts report by the Royal Academy of Engineering and Beauhurst [15].

1.5 Fit for NCCR Muoniverse

The RCA/Imperial modules are designed for TRL 4–6 technologies , technologies far enough along to be pitched to VCs within a week. Student testimonials from the Innovation Challenge confirm this: 'The goal was not to invent a new technology but to develop completely innovative applications by leveraging existing advancements' [4]. DT Prime targets 'pre-commercial' opportunities, but at a stage where £50,000 to £150,000 can generate what Imperial calls 'investible data' , implying the technology has progressed beyond proof of concept [7].

No formal partnership exists between Imperial and Swiss institutions (ETH, EPFL, PSI). The Swiss equivalents are the ETH Pioneer Fellowship programme [18] and the annual DeepTech Investor Summit co-hosted by UBS and ETH Zurich [19]. NCCR Muoniverse's Innovation Platform at HSLU (Hochschule Luzern), led by Christian Hohmann with co-heads Petra Müller-Csernetzky and Shaun West, is the natural adoption point for adapted WAL frameworks [17].

The transferable asset is the methodology , specifically, the technology-first speculative design approach and the WAL toolkit , not the programme structure. NCCR Muoniverse can draw conceptual inspiration from how WAL 'flips' Lean Startup [5][6], but cannot adopt the module as-is because it is institutionally bound to Imperial, designed for MBA and design students, and assumes technologies at higher TRL than muon science currently offers.

Framework 2: Commercial Readiness Level (CRL) , Abbas & Nomvar

2.1 What the Framework Is

The Commercial Readiness Level (CRL) framework is a 9-point scale developed by Professor Ali Abbas and Dr. Mobin Nomvar at the University of Sydney and subsequently commercialized through their company Scimita Ventures, founded in 2017 in Sydney, Australia [11][13]. It is designed to run in parallel with NASA's Technology Readiness Level (TRL) scale and, together with the Manufacturing Readiness Level (MRL) used by the US Department of Defense, forms three supposedly 'orthogonal' or independent axes along which a technology's maturity can be assessed [12].

The CRL scale progresses from recognizing initial commercial opportunities (CRL 1) through measuring market factors (CRL 2), understanding competitiveness (CRL 3), validating the value proposition (CRL 4), validating alignment with market players (CRL 5, identified as a pivotal inflection point), validating regulatory requirements and partnerships (CRL 6), validating financial models and conducting trials (CRL 7), securing customer commitment and validating the scale-up path (CRL 8), and achieving product launch with revenue generation (CRL 9) [11].

The framework's most strategically useful concept is the 'binding constraint.' At any given moment in a technology's development, one of the three axes is the limiting factor that prevents progress toward commercialization. For advanced materials, the binding constraint is typically customer qualification processes; for climate hardware, it is financing institutions; for advanced computing, it is ecosystem trust [12]. Scimita also proposes a synchronization rule: 'the CRL should always equal or exceed the TRL.' When CRL lags TRL, the technology is over-engineered relative to its commercial readiness [11].

2.2 The Critical Documentation Problem

The most significant finding of this assessment concerns the framework's academic standing. Despite being one of the most cited 'commercial readiness' concepts in the deep tech discourse, the Abbas and Nomvar CRL has no peer-reviewed academic paper as its primary documentation. The framework exists as proprietary consultancy IP owned by Scimita Ventures, documented primarily in blog posts on the company's website [11][13] and referenced by third parties including Deep Tech Leaders [12], ITONICS [16], and NITI Aayog (the Government of India's policy think tank) [20]. The most commonly cited reference is an undated institutional document: 'Abbas, A. & Nomvar, M. Commercial Readiness Level (CRL) Framework. University of Sydney' , with no journal, no DOI, no formal publication date [12].

Legend: A DOI (Digital Object Identifier) is a permanent identifier assigned to academic publications. The absence of a DOI means the document is not formally published in the academic record. This does not mean the framework lacks value, but it does mean that any thesis relying on CRL must acknowledge its status as a practitioner framework rather than an academically validated instrument.

This creates a citation challenge for the thesis. The framework is real, influential, and conceptually valuable. But it cannot be cited to a journal article, conference paper, or international standard. Furthermore, unlike NASA's TRL (which has detailed assessment criteria, evidence requirements for each level, and the ISO 16290:2013 standard), the CRL lacks a formal assessment methodology [11]. There is no published evidence checklist per level, no scoring rubric, no independent assessment protocol, and no certification process. Assessment is conducted by Scimita consultants as a proprietary service [13].

An important correction emerged during this assessment: the PDF in the thesis papers folder labeled '22_CRL_Abbas.pdf' is actually an EIT Urban Mobility document describing a different 'Commercialisation Readiness Level' variant , a sales-enablement scoring tool specific to EIT Urban Mobility's project contracts [21]. It is not the Abbas and Nomvar framework and should not be treated as evidence of CRL application.

2.3 Who Has Used It and What Evidence Exists

The most substantial independent application of the CRL framework was found in a policy paper by NITI Aayog (Government of India, 2023) titled 'A New Lens for Innovation in New India' [20]. This paper uses CRL as one of three axes (TRL/CRL/MRL) in what it calls the Techno-Commercial Readiness and Market Maturity (TCRM) Matrix, and applies the matrix to three Indian case studies: UPI (digital payments), Covaxin (vaccine development), and PSLV (space launch vehicle). This is the closest thing to an independent validation of the framework's analytical utility.

Scimita Ventures' own client work includes companies in green hydrogen production (Kinaltek), sulphuric acid processing (Oxleigh Technologies), and non-destructive testing (EcoMine Solutions) [13]. These are consultancy case studies, not published academic validations. No quantitative validation of the framework exists , there are no studies that statistically test whether CRL scores predict commercial success, measure inter-rater reliability, or compare CRL assessments against actual outcomes in a controlled study.

2.4 Competing Readiness Frameworks

The CRL exists in a landscape of competing readiness frameworks. The Investment Readiness Level (IRL), developed by Steve Blank (2013) and operationalized by Village Capital as the VIRAL framework, is investor-centric and tied to funding stages [22][23]. The Manufacturing Readiness Level (MRL), from the US Department of Defense, assesses whether a technology can be produced at scale. The Business Readiness Level (BRL), from KTH Royal Institute of Technology and CNRS, is used in European university incubators [24]. MIT's Dual-Use Readiness Levels separate funding readiness from customer readiness across five parallel axes specifically for defense/commercial dual-use ventures [25]. The ITONICS guide to '14 Readiness Level Frameworks' (September 2025) provides an overview of the landscape [16].

The key distinction: CRL is market-ecosystem-centric (are customers, regulators, financiers ready?), while IRL is investor-centric (what do VCs want to see?) [12][22]. CRL is more descriptive (it tells you where you are),

while IRL is more prescriptive (it tells you what to do next). For deep tech where the market may not yet exist, CRL's descriptive nature is both a strength and a weakness.

2.5 Fit for NCCR Muoniverse

The CRL framework's lower levels (CRL 1–3) are conceptually relevant to TRL 1–3. At CRL 1, a research team can identify that muon tomography might have commercial potential in archaeological dating or battery inspection. At CRL 2, they can begin quantifying the size of the archaeological surveying market. At CRL 3, they can analyze whether competing non-destructive testing technologies pose a competitive threat [11].

However, the operational thinness of these levels limits practical utility. The CRL 1 description , 'recognizing the potential for a commercial opportunity without deep analysis or validation' [11] , applies equally to a muon tomography venture and a mobile app. There are no granular criteria distinguishing what 'recognizing potential' means for a physics technology versus a software product. The framework becomes more useful at CRL 4–5, but by that point the technology has progressed beyond TRL 1–3.

The CRL is best understood as a diagnostic framing tool , a way of making visible the fact that a muon detector can be technically proven (TRL 9) without being commercially viable (CRL 1). This insight is valuable for communicating with stakeholders who may assume that technical excellence implies commercial readiness [12]. But as an actionable assessment instrument for guiding commercialization decisions at TRL 1–3, the framework's proprietary status, lack of peer-reviewed validation, and thin operational criteria limit its utility.

Framework 3: Innosuisse Coaching Methodology

3.1 What Innosuisse Is and Why It Matters

Innosuisse is the Swiss Innovation Agency, the Confederation's primary instrument for promoting science-based innovation. It succeeded the Commission for Technology and Innovation (CTI/KTI) on January 1, 2018, through the Federal Act on the Swiss Innovation Agency (SAFIG Act, SR 420.2) [26]. With an annual budget of approximately CHF 300 million, it operates a portfolio of roughly 20 funding instruments spanning the entire innovation chain [26].

Innosuisse is the most institutionally relevant framework for NCCR Muoniverse because it is the Swiss national framework. Any venture emerging from the consortium will interact with Innosuisse instruments. Understanding exactly what Innosuisse offers, where its instruments are designed to operate, and where the gaps lie relative to muon technology's needs is therefore essential.

3.2 The Three-Phase Start-up Coaching System

Innosuisse's Start-up Coaching is the core support mechanism for science-based startups. It operates as a phased, voucher-based system where startups receive funding for coaching sessions with accredited Innosuisse coaches [27]. The system has three sequential phases.

Initial Coaching provides up to CHF 10,000 over a maximum of 12 months. It serves as the entry point to the system, requiring no prior Innosuisse engagement. A startup works with one accredited lead coach to challenge and review its entire business model, identify strengths and weaknesses, define key milestones, and produce an action plan. This evaluation report becomes the gateway document for admission to Core Coaching [28].

Core Coaching is the main support instrument, providing up to CHF 50,000 in coaching vouchers over a maximum of 36 months [29]. It is milestone-gated: funding is released in tranches based on review of progress against agreed milestones. Each startup has a lead Start-up Coach for strategic guidance plus access to specialist coaches covering intellectual property, financial planning, legal matters, and fundraising. To be eligible, a company must be incorporated in Switzerland within the last 5 years (extended to 10 years for time-intensive life science startups) and have a scalable, defensible business model built on science-based innovation [29].

Scale-up Coaching provides up to CHF 75,000 over 24 months for growth-phase companies [30]. Entry is competitive, via a call rather than automatic progression from Core Coaching. The target is companies that have achieved product-market fit and are preparing for significant scaling, potentially including international expansion.

Legend: The voucher system means Innosuisse does not directly provide the coaching , instead, the startup receives a virtual voucher (like a credit) that it spends on sessions with coaches from Innosuisse's accredited pool [31]. The startup

chooses its own coaches. This design gives startups agency over their support while maintaining quality control through the accreditation system.

3.3 The Coaching Methodology: Custom, Not Codified

A key finding of this assessment is that Innosuisse's coaching methodology is NOT explicitly based on NSF I-Corps or a formal Lean Startup curriculum [31]. It is a custom, mentor-driven model. Coaches are selected for their entrepreneurial experience, not for certification in a particular methodology. Each coaching engagement is bespoke: the content depends on what the startup and coach agree is needed, constrained only by the milestone framework. This is fundamentally different from, for instance, the I-Corps program's standardized 7-week customer discovery immersion where every team follows the same curriculum.

Separately from coaching, Innosuisse funds the Entrepreneurship Training programme operated by venturelab since 2004 [32]. This programme is curriculum-based, covering Business Ideas, Business Concept, Business Validation, and Business Creation, with industry-specific modules for ICT, Life Sciences, Engineering, Energy & Environment, and Social Sciences. Notable alumni include Climeworks, GetYourGuide, and InSphero [32].

3.4 Other Key Instruments

BRIDGE Proof of Concept is jointly operated by Innosuisse and the Swiss National Science Foundation (SNSF). It provides CHF 130,000 over 12 months for researchers to develop research results into marketable products and prepare for a career outside academia [33]. Its approximately 70% founding rate, the proportion of funded researchers who go on to found a startup or spin-off, is the highest conversion rate of any Innosuisse instrument and one of the highest reported by any proof-of-concept program internationally [34]. An independent evaluation of the BRIDGE programme was conducted by econcept AG, which found that 80 startups were created from 121 PoC projects [35].

Innovation Projects are Innosuisse's largest instrument by budget (CHF 140–170 million annually) [36]. Projects with implementation partners follow a 50-50 cost split between Innosuisse and the company, with no financial ceiling. Start-up Innovation Projects offer 70% Innosuisse coverage. The 57% market launch rate within three years post-funding demonstrates effectiveness for the types of ventures these projects serve [34].

3.5 The Evidence Base: Strongest of the Four Frameworks

Innosuisse has the strongest evidence base of any framework in this assessment, consisting of three layers. The Innosuisse Impact Monitor (2021–2023) provides self-assessment outcome data from standardized surveys administered at project completion and again three years post-funding [34]. The KOF/SBFI WTT Monitoring Report (July 2024) provides contextual assessment of the Swiss transfer system [37]. And the KOF

causal study (Hulfeld, Spescha, and Wörter, November 2024) provides independently verified causal estimates [38].

The KOF causal study is particularly important because it addresses the fundamental problem with any evaluation of public funding: do funded firms perform better because of the funding, or would they have performed better anyway because they are already stronger firms? Using a difference-in-differences (DiD) methodology with a matched control group of 537 non-funded firms, analyzing 3,220 firm-year observations across 920 unique firms, the study finds that Innosuisse funding causes an average 20.7% increase in sales and 17.6% increase in employment over five years [38]. The effects grow over time: employment effects start at +7.8% in the year of funding, grow to +17.2% after two years, and reach +27.9% after four years. The effects are strongest and statistically significant specifically for small firms (5–50 employees) and for projects where the ETH Domain or universities of applied sciences serve as the research partner [38].

Legend: Difference-in-differences (DiD) is considered one of the most credible quasi-experimental methods for evaluating program effectiveness. It works by comparing the before-and-after change in outcomes for treated firms versus similar untreated firms. The KOF study addresses the twofold selection problem: firms self-select into applying, and Innosuisse selectively funds the best applicants. The DiD design with matched controls accounts for both [38].

3.6 The Structural Gap for Deep Tech at TRL 1–3

Despite its strong evidence base, a structural mismatch exists between Innosuisse's instrument design and the needs of deep tech ventures at very early TRL [39]. The evidence for this gap is consistent across multiple dimensions.

First, there is no Innosuisse instrument specifically designed for TRL 1–3 commercialization. BRIDGE Proof of Concept is the closest, but its 12-month duration is inadequate for technologies like muon detectors that require multi-year validation cycles [33]. BRIDGE Discovery (CHF 850,000 over 4 years) better fits long-horizon deep tech but targets experienced researchers at faculty level, potentially excluding the postdocs and PhDs who are the likely founders of muon technology ventures [33].

Second, the Core Coaching voucher structure (CHF 50,000) covers advisory services , coach time for strategic guidance, IP strategy, fundraising preparation , but cannot be spent on lab time, detector development, or validation experiments [29]. For a muon technology venture, the binding constraint is not access to business advice but access to research infrastructure and the resources to translate proof of concept into a testable prototype.

Third, the pool of Innosuisse-accredited coaches who have successfully scaled a particle-physics-derived or muon-technology venture is vanishingly small globally [31]. The coaching model assumes that startup-building expertise is transferable across domains. While this is partly true (fundraising, team building, legal structures are domain-general), the specific commercialization pathway for physics technologies requires domain expertise that generic coaches may not provide.

Fourth, Innosuisse's bottom-up model requires firms to take the initiative to apply [26]. This works well for incremental innovation where the path to market is clear, but it may not mobilize the coordinated, mission-oriented effort that frontier physics commercialization requires. The US DOE SBIR/STTR program (\$2.4 million across Phase I and II) and DARPA-style mission programs represent a fundamentally different, top-down approach that may be more appropriate for technologies where the commercial path is not yet visible.

3.7 Interaction with Swiss TTOs

Innosuisse does not operate in isolation. The ETH Domain institutions , ETH Zurich, EPFL, PSI, Empa, Eawag, and WSL , have their own technology transfer infrastructure that interfaces with Innosuisse instruments. ETH transfer, named Tech Transfer Unit of the Year by Global University Venturing Magazine, generates 37–43 spin-offs per year and its spin-offs raised CHF 425 million in 2024 [40]. ETH's total spin-off portfolio is valued at approximately CHF 10 billion [40]. Critically, ETH introduced a policy of 2% equity and a 6–8 week process for spin-off IP licensing , among the most founder-friendly terms in Europe [40]. The ETH Board has published common spin-off licensing principles for all ETH Domain institutions, harmonizing IP practices across Switzerland [41].

EPFL offers the Innogrant (CHF 100,000 for one year of salary for a future entrepreneur developing a spin-off while hosted at an EPFL lab), which complements and can precede Innosuisse Initial Coaching [42]. PSI's interaction with Innosuisse is primarily through Innovation Projects where PSI serves as the research partner. PSI's technologies often require validation timelines that exceed standard project durations, creating a timing mismatch that the thesis interviews (Part 3) should explore [43].

Framework 4: EIC Horizon Europe Deep Tech Support

4.1 What the EIC Is and Why It Matters

The European Innovation Council (EIC) is the European Union's flagship deep tech innovation programme, established under Horizon Europe (2021–2027) with a total budget exceeding €10.1 billion [44]. It was created to address what the EU identified as a structural weakness: Europe produces world-class research but fails to translate it into commercially successful deep tech companies at the same rate as the United States. The EIC's stated mission is to identify, develop, and scale up breakthrough technologies [44].

For NCCR Muoniverse, the EIC matters for two reasons. First, Swiss firms are now fully eligible to participate in all EIC instruments, including equity funding, following the restoration of Switzerland's full association status in Horizon Europe effective January 1, 2025 [45][46]. Second, the EIC's integrated pipeline , covering the entire spectrum from foundational research through commercial scale-up , represents the most comprehensive public deep tech support architecture in Europe, making it a natural reference point for assessing what is available to Swiss ventures beyond national instruments.

4.2 The Four-Instrument Pipeline

The EIC operates through four instruments that form a pipeline intended to cover the entire technology readiness spectrum [44]. EIC Pathfinder funds visionary, early-stage research at TRL 1–4, from basic principles through proof of concept. It succeeds the FET-Open and FET-Proactive programs from Horizon 2020 [47]. Pathfinder operates in two modalities: Open (bottom-up, any field, budget €166 million in 2026) and Challenges (predefined thematic areas, budget €96 million in 2026). Grants typically amount to €3–4 million per project at 100% cost coverage. Projects must be collaborative: a minimum of three independent entities from three different eligible countries [47]. In 2026, Pathfinder Open received 2,103 proposals [47].

EIC Transition bridges the 'valley of death' between Pathfinder (idea/proof of concept) and Accelerator (product/market). It funds the maturation and validation of novel technologies in lab and relevant environments, plus the development of a business case. Target TRL: 3–4 moving toward 5–6. Grants up to €2.5 million. Eligibility is restricted to projects originating from Pathfinder, ERC Proof of Concept, or other EU research programs [48].

EIC Accelerator is the flagship deep tech funding instrument and the one most commonly associated with the EIC brand [49]. It targets TRL 5–8 with a blended finance model combining grants (up to €2.5 million) and equity investments (€0.5–10 million). The application process has three steps: a short proposal (approximately 5 pages), a full proposal (50+ pages), and a remote interview. The end-to-end success rate is 3–7%, making it among the most competitive funding instruments globally [50]. The 2026 budget is €634 million [51].

STEP Scale Up, introduced in 2024, addresses the growth capital gap with pure equity investments of €10–30 million for strategic scaling [52]. Its budget grows from €300 million in 2025 to €900 million over 2025–2027 [52]. This instrument targets companies that have already demonstrated market traction and need capital for expansion, not early-stage ventures.

Legend: The blended finance model (grant + equity) is unique in European public funding [49]. The grant component covers R&D and innovation activities (non-dilutive, meaning the company gives up no equity). The equity component involves the EIC Fund taking an ownership stake in the company, effectively making the EU a shareholder. This combination addresses two gaps simultaneously: the R&D funding gap (covered by the grant) and the growth capital gap (covered by equity). The EIC also provides Business Acceleration Services to all awardees, including coaching, mentoring, corporate/investor matchmaking, and access to the EIC Scaling Club (120 curated deep tech scale-ups) [49].

4.3 The ARPA-Inspired Model and Its Limitations

The EIC is explicitly modeled on the US ARPA/DARPA approach, with dedicated Programme Managers (PMs) who actively manage portfolios rather than passively reviewing proposals [53]. However, academic critiques question whether the EIC has achieved true ARPA-level independence. The German Expert Commission for Research and Innovation (EFI) noted in its 2026 report that 'the EIC is not an independent organisation', the central structural critique [54]. ARPA's effectiveness is widely attributed to its autonomy from political guidance; the EIC, by contrast, reports to the European Commission, which appoints its board and president [54]. Programme Managers are reportedly overloaded, each managing 300 or more projects, compared to DARPA PMs who typically manage 10–15 [55].

4.4 Evidence Base and Its Limitations

The EIC Impact Report 2025 reports €1 billion invested, €2.6 billion co-invested (a 3:1 leverage ratio), 706 startups supported, 52.5% employment growth, 56.6% revenue growth over two years, and 70+ companies reaching 'centaur' status (valuations exceeding €100 million) [36]. The Pathfinder pipeline has produced 1,350+ unique innovations, approximately 400 patents, and 100+ startups established or in formation [36].

However, these figures are programme-self-reported and not independently audited in the publicly available version [36]. No peer-reviewed evaluation of the full Horizon Europe EIC exists yet. The European Court of Auditors Special Report 02/2020 covered only the predecessor SME Instrument, not the EIC in its current form [56]. The CSIL Milan evaluation (2022) used difference-in-differences methodology but covered only the pilot phase [57]. Selection bias and survivorship bias are inherent: the report measures awardees (who were selected through a competitive process) rather than applicants, meaning the reported outcomes may reflect the quality of selected firms rather than the program's causal effect.

4.5 Swiss Participation: From Exclusion to Full Eligibility

Switzerland's participation in the EIC was disrupted from 2021 through 2024. Following the collapse of the Institutional Framework Agreement between Switzerland and the EU, Switzerland was downgraded to third-

country status, meaning Swiss entities could not participate as beneficiaries in Horizon Europe calls and could not receive EIC equity [45]. During this period, SERI (the State Secretariat for Education, Research and Innovation) implemented transitional measures, directly funding 24 Swiss startups with CHF 58 million as 'EIC Accelerator-equivalent' grants and launching the Innosuisse Swiss Accelerator (CHF 86 million) as a domestic substitute [45].

Full association was restored retroactively to January 1, 2025. The Association Agreement was signed in Bern on November 10, 2025 [46]. As of the 2025 programme year, Swiss researchers and innovators participate as full beneficiaries in all Horizon Europe calls, can submit proposals as coordinators, and Swiss startups and SMEs are eligible for both EIC Accelerator grants and equity investments [46].

Early results are promising. In the first round of fully funded Swiss EIC Accelerator beneficiaries (February 2026), four Swiss companies were selected: Neutrality (cybersecurity, Geneva), Verity (AI-powered inventory drones, Zurich), Pregnolia (medical device for pregnancy care, Zurich), and DePoly (chemical recycling, Valais) [58]. In the second round (June 2026), Switzerland accounted for four out of 38 total selections , 'one of the largest groups' per Innosuisse , including BTRY (battery technology, ETH/Empa spin-off, €2.2 million grant) and Nanoflex Robotics (magnetically controlled surgical robots, Zurich, €12.5 million blended finance) [59]. Notably, Swiss firms winning EIC funding are typically alumni of the Swiss startup support pipeline (Venture Kick, Venture Leaders), suggesting that the national ecosystem effectively prepares companies for EU-level competitiveness [60].

4.6 Fit for NCCR Muoniverse

The EIC's primary commercialization instrument , the Accelerator , requires TRL 5 or above [49]. A physics-based venture emerging from NCCR Muoniverse at TRL 1–3 would not be eligible for the Accelerator directly. The relevant pathway for early-TRL physics ventures runs through Pathfinder (TRL 1–4, foundational research) and Transition (TRL 3–6, bridging to validation), both of which require international consortia of at least three entities from three countries [47][48].

The consortium requirement is both a challenge and an opportunity. Building an international consortium adds coordination complexity, but NCCR-type research networks are inherently international , muon physics is conducted collaboratively across PSI, CERN, and international partner institutions. A Pathfinder project on muon technology could potentially Fast Track to the Accelerator full proposal stage once the technology matures, bypassing the initial screening [47].

Strategically, the EIC and Innosuisse are complementary rather than competitive. The EIC offers significantly larger funding (especially through equity, which Innosuisse does not provide) and pan-European investor access [44]. Innosuisse offers a more accessible, less competitive, Switzerland-focused pathway [26]. The logical sequencing for a Swiss deep tech venture would be: Innosuisse instruments for early-stage validation,

EIC Pathfinder/Transition for collaborative research and maturation, EIC Accelerator for scale-up funding, and STEP Scale Up for strategic growth capital.

Proposed Additional Frameworks for Benchmarking

The thesis proposal specifies that four additional frameworks should be selected in consultation with the NCCR Muoniverse team. The following four candidates are proposed based on the literature evidence. Final selection will be agreed before the benchmarking milestone (November 1, 2026).

The first proposed candidate is the NSF I-Corps model and its international adaptations. Hayter et al. (2025) documented how this program has been adopted across eight countries, making it the most internationally validated early-TRL commercialization program [61]. Its core methodology , lean startup combined with intensive customer discovery, delivered through a structured 7-week immersion , directly addresses the need for structured frameworks that work at early stages. As a direct comparator to Innosuisse's bespoke coaching model, I-Corps would provide the thesis with a contrast between methodology-driven and mentor-driven approaches.

The second proposed candidate is CERN Venture Connect (CVC). CERN faces the identical institutional challenge as NCCR Muoniverse: translating fundamental particle physics research into commercial applications [62]. The CVC model , zero equity, express agreements, curated technology portfolio, 60 partner organizations across 18 countries , is the most directly comparable institutional model in the world. Benchmarking against CVC would give the thesis a physics-specific comparator that the other four frameworks lack.

The third proposed candidate is the Augmented Stage-Gate framework (Kiatkaew et al., 2023). This framework integrates TRL and Investment Readiness Level (IRL) into a six-stage, five-gate new product development process. It was validated with three Thai deep-tech case studies covering spin-off, licensing, and non-profit deployment paths [63]. It offers a more structured alternative to the CRL's descriptive approach.

The fourth proposed candidate is the University 16-Mechanism Playbook (Bareš, 2025). This comprehensive policy-oriented framework maps 16 mechanisms covering the entire university-to-market pipeline, from IP frameworks through proof-of-concept finance to exits, with measurable KPIs for each [64]. It could serve as an evaluation rubric for assessing the Swiss ecosystem holistically rather than framework-by-framework.

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